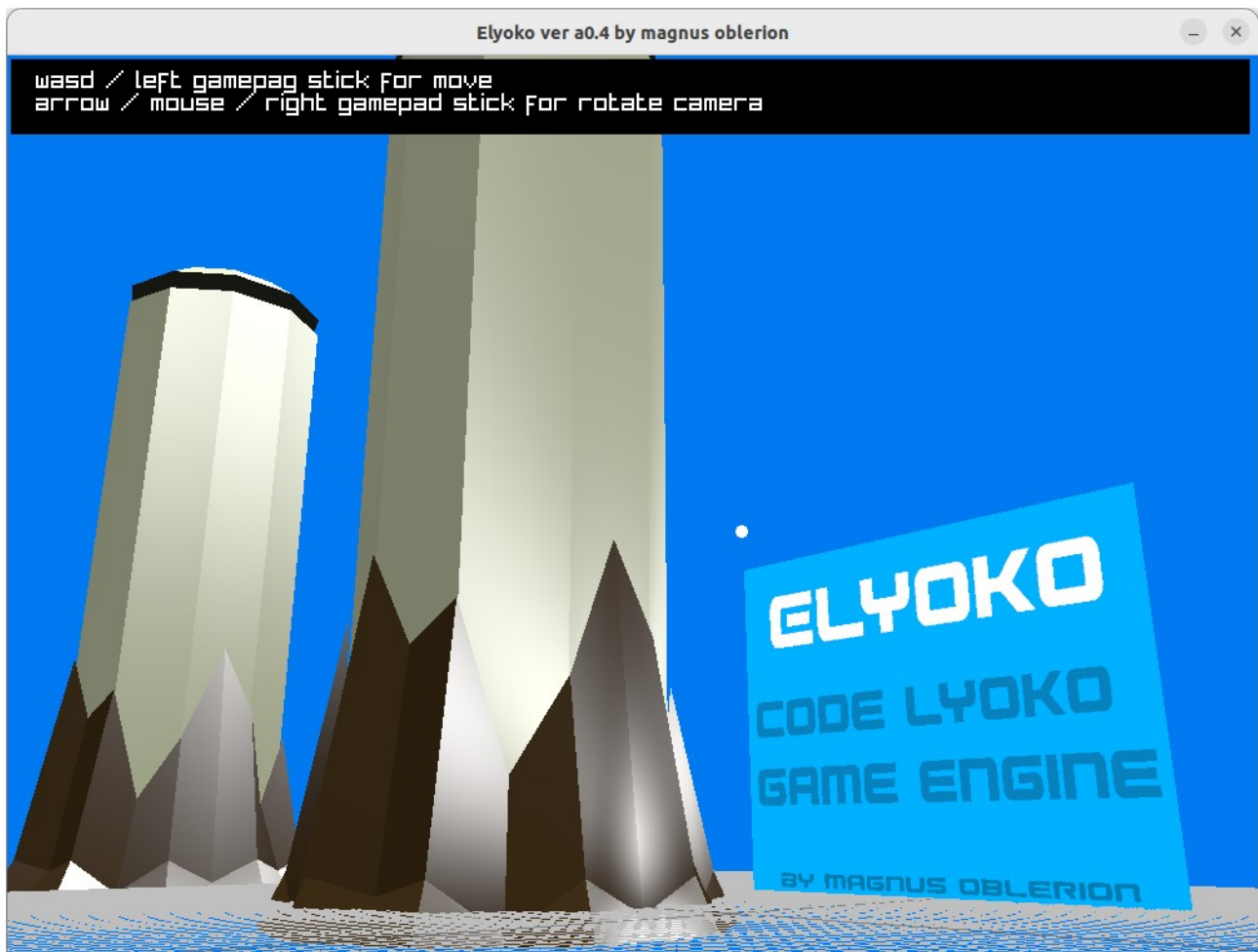


Elyoko

The 3d game engine for
make your lyoko world



by Magnus Oblerion

Description

The 3d game engine make with C , raylib.h and lua.h.

It is lightweight, powerful and easy to extends (lua script).

Features

- Max 100 preload model
- Max 100 preload texture
- Max 100 preload sound
- Max 100 preload music
- load/draw/unload model (obj,gltf,glb)
- load/draw/unload texture (png,jpg)
- load/play sound (wav)
- load/play/pause/stop music (mp3)
- load/draw textured plane from image
- draw 3d primitive (cube,sphere)
- draw 2d primitive (rect,circle,point)
- draw text
- move/rotate camera
- keyboard/mouse input -> camera control
- gamepad input -> camera control

Run script CLI

on window

```
elyoko.exe main.lua
```

on linux

```
./elyoko main.lua
```

Run projet folder CLI

Projet folder need main.lua.

on window

elyoko.exe folder

on linux

./elyoko folder

Entry point

If you create file project.entry beside of binary,
elyoko will run project.lua at start.
Or run project folder if exist.

Your *.lua/folder and *.entry need same name for work.

API

1) Main loop

```
function ELYOK02D()  
-- loop 2d  
end
```

```
function ELYOK03D()  
-- loop 3d  
end
```

2) delta time

```
local dt = deltatime() -- get time between 2 frame
```

3) color

```
local id_color = color(255,0,255,20) -- color r=255,g=0,b=255,a=20  
local id_color = color(255,0,200) -- color r=255,g=0,b=200,a=255
```

4) camera 3d

get camera X

```
local camx = camerax()
```

get camera Y

`local` camy = cameray()

get camera Z

`local` camz = cameraz()

camera move

`cameramove(x, y, z)` -- add value to camera position

camera set position

`camerasetpos(x, y, z)` -- set value to camera position

camera rotate

`camerarotate(rotx, roty, rotz)` -- rotation in radiant

Camera set target

`camerasettarget(x, y, z)` -- set points than camera look

camera lock

`cameralock(state)`

-- state=true player can't move/rotate camera

-- state=false player can move/rotate camera

5) Asset loading

Outside loop or protect for run once ,

if path = icon.png → it try to find it into project folder first before test beside binary.

load model

`loadmodel(path)` -- path is string .obj, .glb

load texture

`loadtexture(path)` -- path is string .png

load plane textured

`loadplanetexture(texture_name.ext)` -- create model plane with texture filename

load sound

`loadsound(path)` -- path is string .wav

load music

`loadmusic(path)` -- path is string .mp3, .ogg

6) Input

btn / btnp

```
if btn(0) then
    -- key w is down or gamepad left stick up
elseif btnp(1) then
    -- key s is pressed or gamepad left stick down
end
```

btn(id) : key is down

btnp(id) : key is pressed

id=0 → key w / gamepad left stick up

id=1 → key s / gamepad left stick down

id=2 → key a / gamepad left stick left

id=3 → key d / gamepad left stick right

id=4 → key x / gamepad button x

id=5 → key c / gamepad button a

mouse

```
local x,y,btnl,btnm,btnr = mouse()
-- x = x mouse
-- y = y mouse
-- btnl = true/false mouse left button
-- btnm = true/false mouse mid button
-- btnr = true/false mouse right button
```

7) Sound

Play

playsound(name) -- name = string , filename without path

Set volume

setsoundvolume(name, vol) -- name = string , vol = (0.0 → 1.0)

Set pan

setsoundpan(name, pan) -- name = string , pan = (0.0 → 1.0)

Set pitch

setsoundpitch(name, pitch) -- name = string , pitch = (0.0 → 1.0)

8) Music

Play

playmusic(name) -- name = string , filename without path

Pause

pausemusic(name) -- name = string

Stop

stopmusic(name) -- name = string

Set volume

setmusicvolume(name, vol) -- name = string , vol = (0.0 → 1.0)

Set pan

setmusicpan(name, pan) -- name = string , pan = (0.0 → 1.0)

Set pitch

setmusicpitch(name, pitch) -- name = string , pitch = (0.0 → 1.0)

9) 2d

work only in ELYOKO2D

draw pixel

pix(x, y, color())

draw fill rectangle

rect(x, y, width, height, color())

draw line rectangle

rectb(x, y, width, height, color())

draw fill circle

circle(x, y, radius, color())

draw line circle

circleb(x, y, radius, color())

draw text

text(str, x, y, scale, color())

unload texture

```
deltexture(name.ext)
```

draw texture

```
drawtexture(name.ext, x, y)
```

10) 3d

work only in ELYOKO3D()

draw cube

```
cube(x, y, z, width, height, depth, color())
```

draw sphere

```
sphere(x, y, z, radius, color())
```

unload model

```
delmodel(name.ext) -- name.ext is name without path
```

draw model

```
drawmodel(name.ext, x, y, z, rot_x, rot_y, rot_z, scale)  
-- rotation in radian  
-- scale > 0 , float multiply scale of model
```

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